



A3 Cell structure

A · Unity and diversity · IB Diploma · Concept guide and original exam-style practice

Understand the concept

All living organisms are composed of cells, the basic structural and functional unit of life, which are either prokaryotic (no nucleus or membrane-bound organelles, small, circular DNA) or eukaryotic (membrane-bound nucleus and organelles such as mitochondria, ER, Golgi apparatus and lysosomes, allowing compartmentalisation of function). The endosymbiotic theory proposes that mitochondria and chloroplasts originated as free-living prokaryotes engulfed by an ancestral eukaryotic cell, supported by evidence such as their double membranes, own circular DNA and 70S ribosomes, and independent division by binary fission.

Worked example: follow the reasoning

An electron micrograph shows a mitochondrion with a measured length of 20 mm on the image; the image was taken at a magnification of $\times 8000$. Step 1: actual size = image size $\div$ magnification = $20 \text{ mm} \div 8000 = 0.0025 \text{ mm}$. Step 2: convert to more convenient units: $0.0025 \text{ mm} \times 1000 = 2.5 \mu\text{m}$. Conclusion: the real mitochondrion is approximately $2.5 \mu\text{m}$ long, consistent with typical mitochondrial dimensions ($1\text{--}10 \mu\text{m}$).

Quick check · multiple choice

Which organelle is the main site of ATP synthesis by oxidative phosphorylation in a eukaryotic cell?

- A Golgi apparatus
- B Mitochondrion
- C Free ribosome
- D Lysosome

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student examines a plant cell using both a light microscope and an electron microscope. [6]
- (a) Distinguish between magnification and resolution. [2]
 - (b) Explain why an electron microscope can resolve structures such as ribosomes that are not visible with a light microscope. [2]
 - (c) State two pieces of structural evidence supporting the endosymbiotic origin of chloroplasts. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) Magnification is how many times larger an image is than the actual object; resolution is the minimum distance between two points at which they can still be distinguished as separate.
- (b) Electron microscopes use a beam of electrons, which has a much shorter wavelength than visible light; resolution is limited by wavelength, so the shorter wavelength of electrons allows much smaller structures (such as ~20–30 nm ribosomes, below the ~200 nm resolution limit of light) to be distinguished.
- (c) Any two of: chloroplasts have a double membrane (consistent with engulfment by phagocytosis); they contain their own circular DNA, separate from the nucleus; they contain their own 70S ribosomes similar to bacterial ribosomes; they can divide independently of the cell by a process similar to bacterial binary fission.

Quick check answer

B. The mitochondrion houses the electron transport chain and ATP synthase on its inner membrane, generating most of the cell's ATP via oxidative phosphorylation.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.